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A Comprehensive Framework for Multiphase Pump Design and Optimal Operation in Upstream Applications

Padmanaban Manokaran, Senior Engineer- Projects, ADNOC Al Dhafra Petroleum, Abu Dhabi, UAE; Fahad Al Rumaithi, Project Manager, ADNOC Al Dhafra Petroleum, Abu Dhabi, UAE; Sinan Mohammad Abdelhadi, Senior Project Manager, ADNOC Al Dhafra Petroleum, Abu Dhabi, UAE; Taha Abdulla Al Jabri, VP Operations and Projects, ADNOC Al Dhafra Petroleum, Abu Dhabi, UAE

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Abstract

The oil and gas industry continues to face mounting challenges in developing marginal fields and optimizing production from mature assets, where conventional separation and pumping systems often prove economically prohibitive or technically impractical. Multiphase pumps (MPPs) have emerged as a transformative technology, enabling operators to overcome these barriers by efficiently handling untreated well streams with varying gas-liquid ratios and solids content. This paper presents a holistic, field-tested framework developed through Al Dhafra Petroleum's extensive operational experience with multiphase pumping systems. The methodology integrates rigorous selection criteria, reliability-focused design principles, manufacturing excellence, advanced monitoring technologies, and best operational & maintenance practices to address prevalent challenges such as mechanical failures, performance degradation, and suboptimal system availability.

Implementation of this structured approach has yielded demonstrable improvements, including complete elimination of seal failures, sustained pump availability exceeding 95%, and substantial reductions in maintenance expenditures. Beyond immediate operational benefits, the framework provides upstream operators with a replicable methodology for deploying multiphase pumping technology across diverse field conditions, particularly in challenging environments such as high-GOR reservoirs, remote locations, and marginal field developments. The paper concludes with actionable recommendations for asset teams seeking to enhance production efficiency while minimizing life-cycle costs through optimized multiphase pumping solutions.

Introduction

Background

Al Dhafra Petroleum Operations Company Limited (ADNOC), a joint venture with KADOC established in late 2013, holds onshore concession areas in Abu Dhabi, UAE. The company currently operates the Haliba

field, which serves as a central production hub for hydrocarbons from Haliba and adjacent satellite fields. Produced well-fluids are routed to the Haliba Central Gathering Facilities (CGF) via Multi-Port Selector Valves (MPSVs). These valves enable selective routing from multiple wells (up to seven per MPSV), optimizing flow control and testing operations. At the CGF, fluids are boosted using Multi-Phase Pumps (MPPs) and transferred through a ~65 km pipeline to ADNOC Onshore ASAB facilities for downstream processing and export.

To address metering uncertainties associated with Multi-Phase Flow Meters (MPFMs), Custody Transfer Metering Facilities (CMF) have been installed at Haliba CGF. These facilities, operated by Oilserv, include a 3-phase separator, oil booster pump, and metering skid, ensuring accurate phase separation and fiscal measurement before recombination at the MPP suction for onward transfer. This infrastructure enables Al Dhafra Petroleum to leverage existing ASAB processing capabilities while maintaining operational efficiency and measurement integrity in its production network.

The global energy landscape continues to evolve, with operators facing increasing pressure to maximize recovery from existing assets while developing more challenging reservoirs in cost-effective ways. In this context, multiphase pumping technology has transitioned from a niche solution to a mainstream production enhancement tool, particularly for marginal fields, deepwater developments, and high-GOR applications where traditional separation infrastructure proves economically unviable.

Despite their demonstrated benefits, multiphase pump installations frequently underperform due to fundamental mismatches between pump capabilities and actual operating conditions. Industry data suggests that nearly 40% of premature MPP failures stem from improper pump selection, while another 35% relates to operational practices that fail to account for dynamic production characteristics. These challenges are compounded by the mechanical complexity of multiphase pumping systems, which must accommodate wide variations in gas volume fraction (GVF), solids content, and flow regimes while maintaining operational reliability.

This paper presents a comprehensive methodology developed through Al Dhafra Petroleum's seven-years operational experience with twin-screw multiphase pumps across multiple field developments. The framework addresses the complete lifecycle of multiphase pumping systems - from initial selection through long-term operation - incorporating lessons learned from both successes and failures. By combining systematic technical evaluation with practical field implementation strategies, the approach provides operators with a proven pathway to optimize multiphase pumping performance while mitigating common failure modes.

Objectives and Scope

The primary objective of this paper is to present a comprehensive, field-tested framework for the selection, design, manufacturing and operation of multiphase pumping (MPP) systems in the oil and gas industry. This paper aims to address the following:

- Development and validation of a structured methodology for deploying multiphase pumps, based on Al Dhafra Petroleum's operational experience.
- Integration of selection criteria, reliability-centered design principles, advanced monitoring technologies, manufacturing, testing requirements and best operational & maintenance practices into a unified framework.
- Mitigation of common operational challenges, including mechanical failures, performance degradation, and low system availability.
- Application of the framework to diverse and challenging environments such as high-GOR (gas-oil ratio) reservoirs, remote field locations, and marginal developments.
- Provision of actionable recommendations for asset teams to enhance production efficiency and reduce life-cycle costs using optimized multiphase pumping solutions.

Multiphase Pumps: Fundamentals and Applications

A **Multiphase Pump (MPP)** is an advanced turbomachinery solution engineered to transport a heterogeneous mixture of **oil, gas, water, and entrained solids** (e.g., sand, fines) in a single-phase stream **without prior separation**. It enables production from **remote, marginal, or subsea reservoirs** by eliminating the need for conventional **separators, compressors, and phase-specific pipelines**, thereby streamlining infrastructure and reducing operational complexity for remote, marginal, or offshore fields where infrastructure is limited.

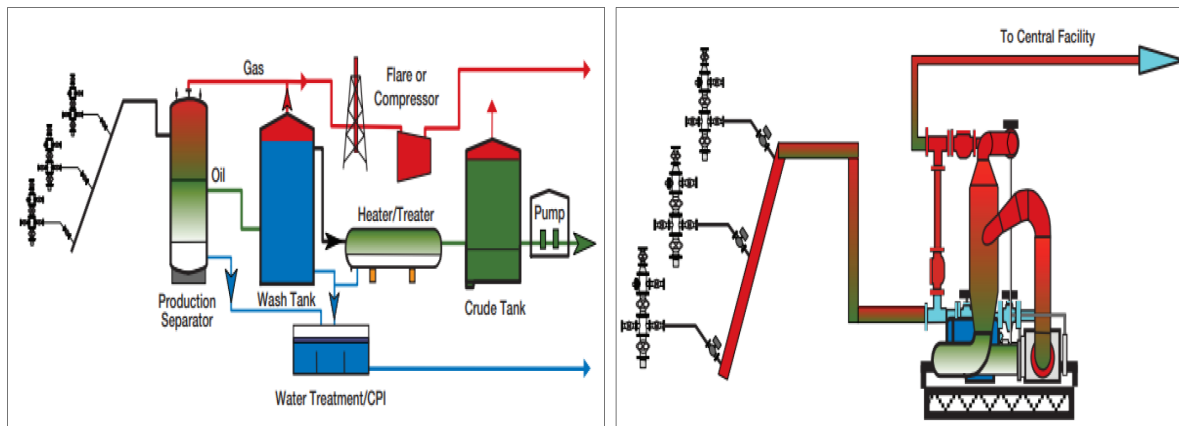


Figure 1—Representation of conventional Separation versus Multiphase Pumping System

Operating Principles of MPPs

Multiphase pumps impart mechanical energy to mixed-phase fluids via rotating elements. Two primary design architectures are employed:

Helico-Axial (Centrifugal): Each stage consists of a rotating impeller and a static diffuser. The impeller imparts kinetic energy to the fluid, which is converted into pressure energy by the diffuser. The helico-axial geometry minimizes radial flow components, maintaining a stable axial flow and preventing phase separation.

Twin-Screw (Positive Displacement): They operate by using two intermeshing helical screws that rotate inside a close-fitting casing. As the screws rotate, they form sealed cavities that move fluid axially from the suction to the discharge end. The screws are synchronized by timing gears, ensuring no metal-to-metal contact, which reduces wear and allows for dry gas handling. These pumps are self-priming and maintain a constant volumetric flow, regardless of pressure variations.

Core Components of a Multiphase Pump Skid

A multiphase pump skid is a modular, integrated system designed to transport mixed-phase fluids oil, gas, water, and solids without prior separation. Its components are engineered for high reliability, compactness, and adaptability to remote or marginal field conditions.

Table 1—Overview of Multiphase Pump Components, System Packages, and Functions

Component	Function
Pump (Rotor & Housing)	Transfers energy to the fluid stream
Motor (Electric/Diesel)	Drives the pump via shaft or magnetic coupling
Variable Frequency Drive (VFD)	Modulates motor speed for flow and pressure control
Seal Gas System	Maintains pressure barrier to prevent seal leakage
Lube Oil System	Cools and lubricates bearings
Suction/Discharge Piping	Interfaces with wellhead and export systems
Instrumentation	Real-time monitoring of pressure, temperature, vibration, and flow
Control System (PLC/HMI)	Automates operations, safety interlocks, and remote-control integration

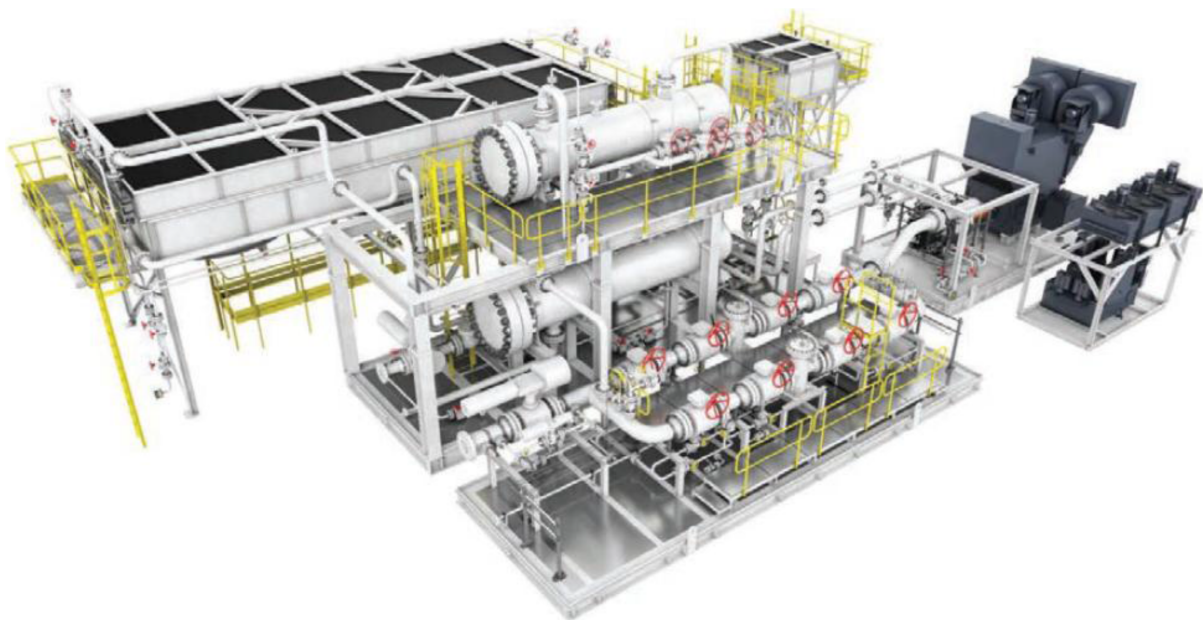


Figure 2—System Overview of a standard Twin-screw pump and package

Performance Advantages

Multiphase pump skids are highly versatile systems designed to handle mixed-phase flows of oil, gas, and water without prior separation. They are especially valuable in remote onshore marginal fields, where infrastructure is limited and production economics are tight. For high gas-to-oil ratio (GOR) wells, these pumps maintain flow assurance and prevent gas locking, ensuring stable operation. Additionally, MPP skids are widely used in subsea tiebacks and booster stations, where compact, high-reliability systems are needed to transport multiphase fluids over long distances with minimal intervention

Table 2—Key Advantages Multiphase vs Conventional Pumping Systems

Benefit	Description
CAPEX Reduction	Eliminates need for separators, compressors, and phase-specific piping
Accelerated Monetization	Enables early production without full-scale facility buildout
Space Optimization	Compact skid design suitable for constrained environments
Enhanced Production	Sustains flow from declining or high-GOR wells
Improved Reservoir Sweep	Reduces wellhead backpressure, enhancing recovery efficiency

Key Challenges in Multiphase Pumping Applications

The successful implementation of multiphase pumping (MPP) systems hinges on a comprehensive understanding of the operational complexities associated with untreated well streams and harsh field conditions. These challenges span four critical domains:

Selection Complexity and Design Constraints

The absence of universally accepted selection standards often results in mismatched pump configurations. Key oversights include:

- **Highly variable fluid properties:** Gas Volume Fractions (GVF) ranging from 0–95%, fluctuating viscosities, and unpredictable solids content.
- **Unsteady flow regimes:** Severe slugging and pressure surges, particularly in long tieback or marginal field applications.
- **Mismatch between test and field conditions:** Laboratory trials typically use clean, stable fluids, which fail to replicate the dynamic multiphase behaviour observed in real-world operations.

Industry data indicates that approximately **60% of MPP underperformance** stems from inadequate front-end engineering and selection-phase assumptions.

Mechanical Integrity and Reliability Risks

Field reliability studies highlight three dominant failure modes:

- **Seal system degradation** (42% of failures): Triggered by gas ingress during low-flow or transient operations and exacerbated by frequent start-stop cycles.
- **Bearing and shaft fatigue** (31%): Attributed to unbalanced radial loads in twin-screw pumps and axial thrust fluctuations in helico-axial designs.
- **Erosive wear** (27%): Common in high-sand-content wells, where fluid velocities exceed material erosion thresholds.

These failure mechanisms underscore the need for robust mechanical design and material selection tailored to multiphase service.

Performance Degradation and Efficiency Losses

Long-term operational data reveals several performance-limiting phenomena:

- **Gas locking:** Occurs when GVF exceeds 85%, leading to volumetric efficiency losses of 30–50%.
- **Progressive efficiency decay:** Typically, 2–3% annually, due to wear in critical internal clearances.
- **Hydraulic instability:** Transient flow conditions induce vibrations that often exceed ISO 10816 vibration severity thresholds, risking premature failure.

Operational and Environmental Constraints

MPP systems deployed in remote or offshore environments face additional constraints:

- **Limited access for maintenance:** Demands high-reliability, low-maintenance designs.
- **Corrosive service conditions:** Elevated H₂S and CO₂ levels necessitate corrosion-resistant alloys and advanced sealing technologies.

Systematic Selection Criteria for Multiphase Pumping Technologies

Multiphase flow is inherently complex due to the co-existence of gaseous, liquid, and sometimes solid phases within a single pipeline. These phases interact in unpredictable ways, leading to variable flow regimes such as slug, annular, bubble, or stratified flows. The behavior of multiphase mixtures is influenced by factors like pressure, temperature, fluid composition, flow velocity, and pipeline geometry. The adoption of systematic selection criteria is not merely a matter of protocol but a foundational step toward achieving safety, reliability, efficiency, and overall project success.

Technical Comparison of Twin Screw vs. Helico-Axial

In multiphase production systems, pump selection is critical to ensure reliable transport of mixed-phase fluids under varying flow regimes, gas volume fractions (GVF), and field conditions. The two dominant technologies Twin Screw Pumps (TSP) and Helico-Axial Pumps (HAP) offer distinct advantages and limitations.

Twin Screw Pumps (TSP). TSPs are positive displacement rotary pumps built to API 676 standards. They feature two intermeshing helical screws driven by timing gears, forming sealed chambers that transport fluid axially from suction to discharge. Key attributes include:

- **Wide GVF Handling:** Capable of managing GVF up to 98% with a liquid management system to prevent dry running and overheating.
- **Low-Speed Operation:** Typically, ≤ 3600 RPM, reducing wear and allowing use of cost-effective drivers.
- **High Viscosity Tolerance:** Ideal for heavy oil and slug flow conditions.
- **Seal Integrity:** Double mechanical seals with Plan 53B/54 flushing systems.
- **Operational Flexibility:** Tolerant to transient flow and suction pressure variations.
- **Limitations:** Poor solid handling due to tight internal clearances; requires pressure limiting valves and careful upstream flow splitting in parallel configurations.

TSPs are globally the most installed multiphase pump type, with vendors including ITT-Bornemann, Flowserve, Leistritz, and Warren-Colfax.

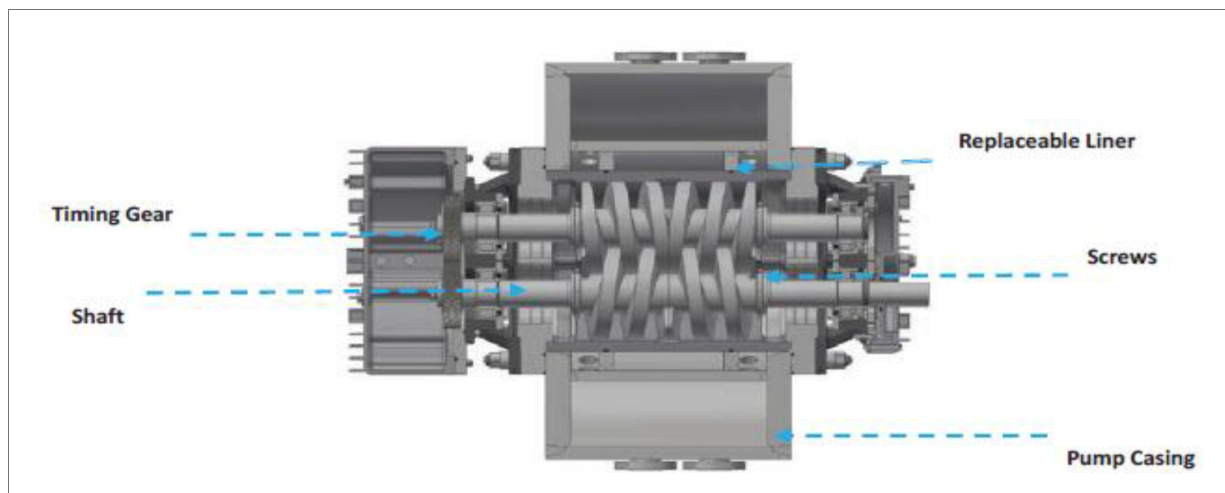


Figure 3—The cross-sectional view of a typical Twin-screw pump

Helico-Axial Pumps (HAP). HAPs are rotodynamic axial flow pumps designed per API 610 with multiphase-specific exceptions. They use stacked helical impellers and axial diffusers to maintain flow homogeneity and minimize gas-liquid separation. Key features include:

- **Axial Flow Design:** Prevents centrifugal separation of gas and liquid phases, maintaining mixture integrity.
- **Solid Tolerance:** Large clearances and open impeller geometry resist sand erosion.
- **Buffer Tank Integration:** Smooths transient slug flow and stabilizes shaft torque fluctuations.
- **Cartridge Design:** Facilitates easy maintenance and reduced downtime.
- **Limitations:** Requires high operating speeds (up to 6800 RPM), buffer tank sizing, and significant recirculation ($\geq 40\%$ rated flow) during low-flow conditions.

Main vendors include Sulzer and Framo.

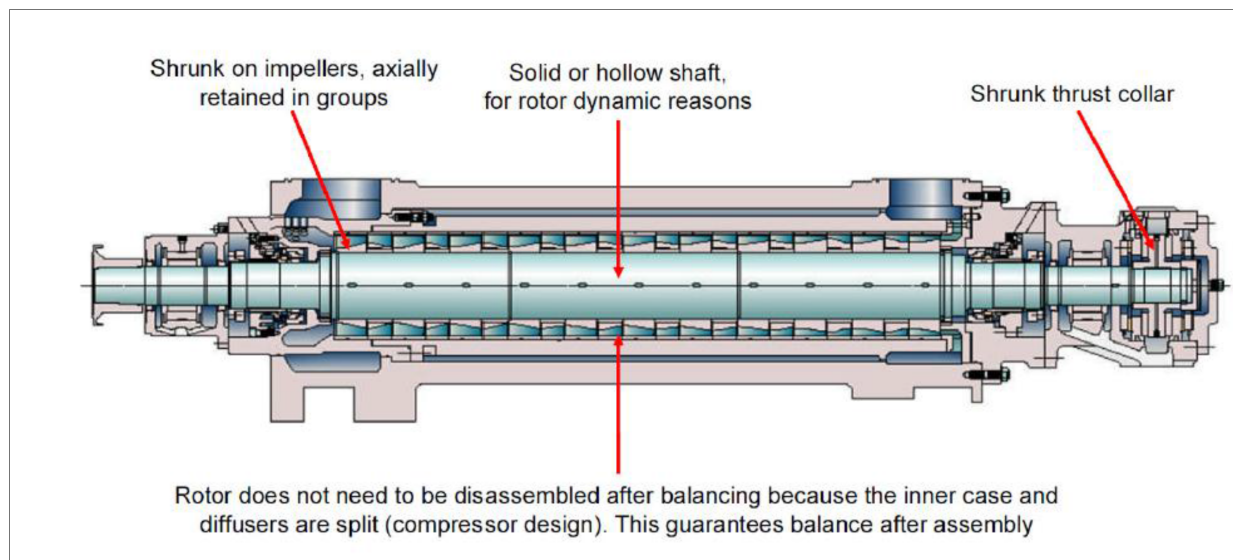


Figure 4—The cross-sectional view of a typical Helico axial pump

Key Technical Criteria for Selecting TSP vs. HAP

Two leading technologies Twin Screw Pumps (TSP) and Helico-Axial Pumps (HAP) offer distinct advantages and are suited to different application scenarios. This section outlines the key technical criteria to consider when selecting between TSP and HAP, enabling organizations to make systematic, informed decisions. The selection of TSP versus HAP for multiphase pumping systems is dictated by a balanced evaluation of fluid properties, operating conditions, reliability, integration needs, environmental compliance, and economics.

- TSPs are ideal for high-GVF, high-viscosity, and remote operations where reliability, flexibility, and long-term efficiency are critical.
- HAPs are better suited for applications with solid-laden flows and where ease of maintenance is prioritized, but they require more operational safeguards and have higher wear rates.

This quantitative approach reduces selection bias and ensures the chosen pump type (twin-screw vs. helico-axial) aligns with actual operating conditions. This comparative framework supports informed decision-making based on field conditions such as GVF levels, viscosity, solids content, and operational flexibility. Systematic criteria ensure that the chosen technology aligns with project objectives, regulatory

landscapes, and sustainable operational practices delivering value and performance across the asset lifecycle.

Table 3—Technical Comparison of Twin Screw Pump (TSP) and Helico-Axial Pump (HAP) for Selection

Criteria	Twin Screw Pump (TSP)	Helico-Axial Pump (HAP)
Operating Speed	Low speed (≤ 3600 RPM) → lower wear and better longevity	High speed (≤ 6800 RPM) → increased wear and maintenance
Turndown Range	Wide (10–100%) without recirculation	Limited ($\geq 40\%$ BEP) → requires buffer tank and recirculation
GVF Handling	Sustained 100% GVF with minimal liquid flush (2% rated flow)	Limited duration at 100% GVF → needs larger buffer and more recirculation
Volumetric Efficiency	High efficiency at elevated GVF and viscosity	Lower efficiency at high GVF
Viscosity Tolerance (Liquid Slugs)	More efficient at high viscosity	More efficient at low viscosity
Viscosity Tolerance (Gas Slugs)	Less efficient at low gas viscosity	More efficient at low gas viscosity
Solid Handling Capability	Less tolerant to solids	More tolerant due to open impeller and axial flow path
Flow Pulsation	Pulsating flow → requires dampening	Steady flow → no dampening needed
Capacity Upgrade Flexibility	Easy via screw pitch, diameter, and gear modifications	Limited to impeller diameter and speed changes
Impeller Arrangement	Integral screw design	Keyed impellers → prone to wear over time
Maintenance Design	No cartridge system	Cartridge barrel → easier maintenance and reduced downtime
Hydraulic Balance	Achieved via double suction and centre discharge	Requires balancing drum → risk of plugging and imbalance
Shaft Design	Short, rigid → less vibration	Long, flexible → more vibration and deflection
Wear and Wearable Parts	More wearable parts but lower wear rate due to low speed	Fewer parts but higher wear rate due to high speed
Mechanical Losses	Higher due to more seals, bearings, and gear sets	Lower due to fewer mechanical interfaces
Minimum Flow Power Requirement	Lower	Higher

Strategic Engineering Framework for Optimized Multiphase Pump System

Designing a high-performance, field-adapted MPP system demands a structured engineering approach that integrates hydraulic analysis, mechanical integrity, instrumentation, control logic, and safety systems. This ensures operational reliability, especially in challenging environments such as remote, marginal, or high-temperature fields.

Front-End Design Considerations

- **Production Envelope:** Define key operating ranges for flow, pressure, temperature, GVF, and solids; consider evolving production profiles for long-term adaptability.
- **Fluid Characterization:** Analyze fluid properties (viscosity, density, emulsions, wax/paraffin) and quantify solids to assess erosion risk and guide material choices.
- **System Objectives:** Clearly state operational goals (e.g., pressure boosting, early separation) and specify the duty cycle (continuous or batch operation) to match system design with requirements.
- **Design Turndown:** Design for a 3:1 turndown ratio to handle variable flow. Employ VFDs and automated controls to respond to changing conditions.

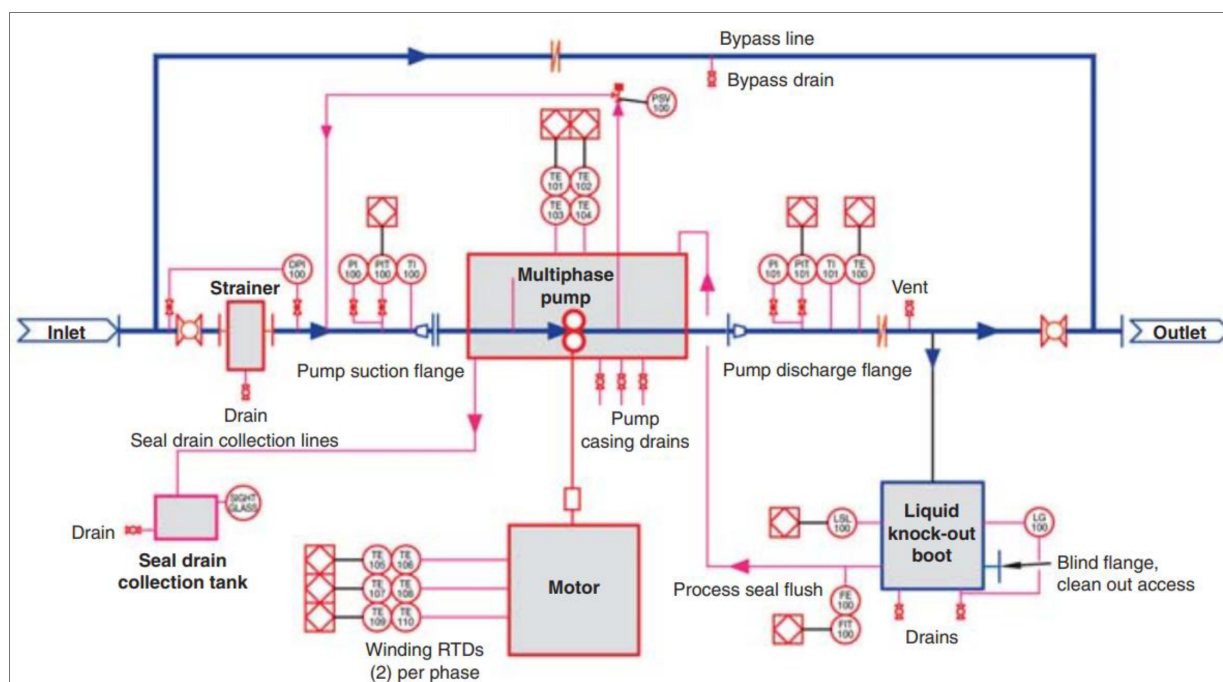
Pump Selection

- The appropriate pump type such as twin-screw, helico-axial, or progressive cavity pumps should be chosen based on gas volume fraction (GVF), fluid viscosity, and required flow rates as described in above section-4.

Integrated Engineering and Safety Strategy

- **Mechanical and Process Engineering:**
 - Comprehensive mechanical and process design should begin with detailed pump duty point evaluation and pump curve analysis across the full gas volume fraction (GVF) spectrum.
 - Material selection for critical components such as pump casings and rotors must prioritize corrosion and erosion resistance using corrosion-resistant alloys (CRAs) like Duplex Stainless Steel or Inconel, supplemented by wear-resistant coatings when necessary.
 - Effective thermal management strategies should account for ambient conditions exceeding 50°C.
 - Skid layouts must optimize spatial efficiency, ensuring modularity, vibration isolation, clear maintenance access, and the ability to accommodate redundant or parallel pump configurations for enhanced operational reliability.
 - System protection should incorporate pressure safety valves (PSVs) on both suction and discharge lines, a bypass or recirculation loop with flow control for low-demand scenarios, and a downstream backpressure control valve to maintain stable differential pressure.
- **Instrumentation & Controls:** High-fidelity instrumentation is necessary for precise monitoring and control, including suction and discharge pressure transmitters, flow transmitters, seal pressure and differential sensors, bearing and oil temperature sensors, and vibration monitors. Control architecture should feature variable frequency drive (VFD) regulation, closed-loop control for pressure and flow, and comprehensive safety interlocks. Automated shutdown logic must respond to critical conditions such as high vibration, overpressure, or seal failure to ensure safe and stable multiphase pump (MPP) operation under dynamic field conditions.
- **Electrical Design:** Motors must be sized to 110–120% of the maximum anticipated load, with VFDs providing soft start capabilities, torque control, and harmonic mitigation. Electrical design must include robust cable routing compliant with IEC or NEC hazardous area classifications, as well as dedicated grounding and bonding systems to minimize static discharge risks. Separation of control and power cable trays is essential to prevent electromagnetic interference (EMI).
- **Control System:** Integration of a local human-machine interface (HMI) with remote SCADA/RTU systems is crucial for operational visibility and control. Control logic should provide surge protection, suction loss detection, emergency shutdown (ESD) capability, and include SIL-rated components for all critical safety and shutdown loops.
- **Hydraulic Design:** Suction and discharge piping must be engineered with long-radius elbows and minimal turbulence-inducing features upstream of the pump. Effective backpressure management and upstream sand/slug mitigation using buffer volumes, slug catchers, or desanders are essential to prevent flow instability and protect pump integrity.
- **Safety and Compliance:** The design must undergo rigorous hazard and operability analysis (HAZOP) and Safety Integrity Level (SIL) assessment. Emergency shutdown systems must be robust, and all designs must conform to applicable industry codes, such as API 676/610/682, ASME B31.3/16.5/Section VIII, IEC/NEC, and NACE MR0175, to ensure comprehensive safety and regulatory compliance.

- 3D Modeling and Layout Engineering:** All system components should be integrated within a 3D plant modeling environment (e.g., AutoCAD Plant 3D, PDMS, AVEVA E3D), enabling early clash detection and alignment with site plot plans, foundation layouts, and cable routing schemes. Maintenance access requirements should be explicitly addressed, including space for equipment removal, lifting tools, scaffolding, and safe operator access and visibility zones, with adequate lighting considered in the design.
- Documentation:** The project deliverables must comprise comprehensive engineering documentation, including detailed data sheets, P&IDs with control logic annotation, general arrangement and layout drawings, instrument hook-up and loop diagrams, electrical single-line diagrams and MCC layouts, and complete hazard analysis reports, supporting both successful implementation and long-term operability of the MPP system.



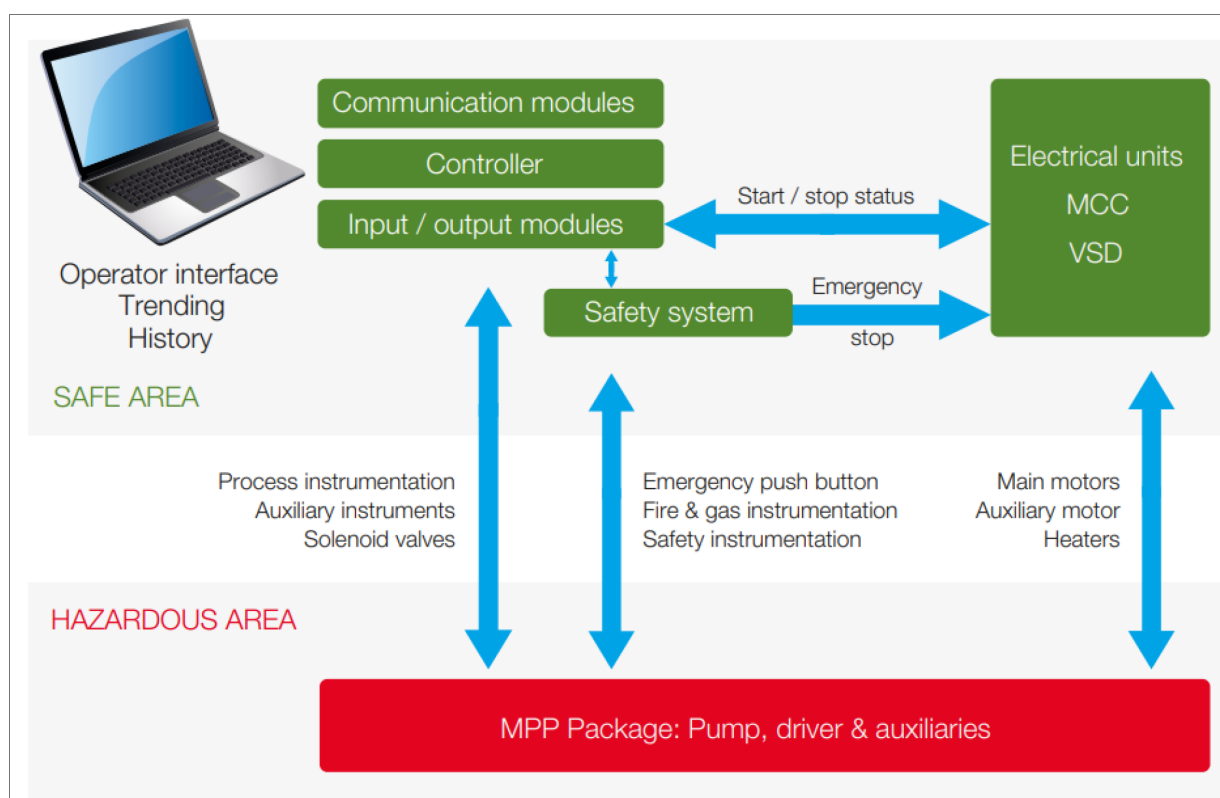


Figure 5a—Illustration of the Monitoring and Control Processes for the MPP Package and Related Systems

Setting Stringent Performance Criteria

Multiphase Pump (MPP) performance testing is a critical step in ensuring equipment meets design criteria, operational reliability, and safety standards before deployment. A comprehensive performance test validates hydraulic behavior, mechanical integrity, control system response, and instrumentation accuracy under representative conditions. These tests are conducted both at the factory and at the field site.

Types of Performance Tests. Below is a categorized list of performance tests commonly used in engineering and industrial systems, particularly for multiphase pumping and surface facility design.

Table 4—Multiphase Pump Performance Tests

Test Type	Location	Purpose
Hydrostatic Test	Factory	Verify pressure containment integrity
Mechanical Run Test	Factory	Ensure rotor balance, temperature stability
Performance Test	Factory/Test Loop	Confirm flowrate, pressure, power, efficiency
Control System Test	Factory	Verify logic, alarms, trips, VFD operation
Integrated Factory Acceptance Test or Complete Unit Test (CUT)	Factory	Conduct a system integration check involving the motor, lube oil system, seal oil system, piping skid, separator skid, VFD, and unit control panel for the project equipment, rather than the factory equipment, to ensure verification under simulated field conditions.
Field Commissioning Test	Site	Validate integration with field piping and control
Site Performance Test	Site	Validate actual performance under field conditions

Key Performance Parameters and Acceptance Criteria. A concise summary of Key Performance Parameters and Acceptance Criteria for multiphase pumping systems, designed for marginal field use:

Table 5—Multiphase Pump Performance Criteria

Parameter	Typical Acceptance Range
Flowrate	Within $\pm 5\%$ of rated at design point
Differential Pressure	Within $\pm 3\%$ of rated differential at specified flowrate
Power Consumption	Within $\pm 5\%$ of rated motor kW at design point
Vibration (RMS Velocity)	≤ 5 mm/s (ISO 10816-3 for rotating equipment)
Temperature Rise	Bearing/oil temp $\leq$ OEM design limits
Seal Gas Flow	Consistent and within OEM range
GVF Tolerance (if tested)	Maintain pump head up to 80% GVF (if test rig supports)

Smart Manufacturing and Qualification Protocols for Reliable Multiphase Pump System Delivery

The deployment of Multiphase Pump (MPP) systems within complex oil and gas applications demands a rigorously engineered approach spanning design, fabrication, quality assurance, and site integration. This section presents a systematic methodology for MPP delivery, emphasizing advanced manufacturing protocols, compliance with international codes, and lessons learned through project lifecycle management. The paper provides actionable guidance for EPC contractors and operators seeking to elevate system reliability, minimize commissioning risks, and accelerate time-to-production in challenging environments. Particular focus is given to traceability, digital integration (including digital twin updates), and harmonization with global standards (API, IEC, NACE, ASME) to ensure enduring performance and regulatory alignment.

Procurement & Vendor Qualification

- **AVL Compliance:** Utilize pre-qualified manufacturers with proven API certification, robust QMS, audited performance history and proven delivery records. Execute detailed factory audits, capability assessments, and risk analysis aligned with project specifications.
- **Traceability:** Mandate full traceability (heat numbers, MTCs) for all critical item's rotors, casings, seals to NACE MR0175/ISO 15156 requirements for sour service
- **Third-Party Inspection:** Appoint accredited agencies (e.g., TUV, Lloyd's, Bureau Veritas) to independently witness fabrication milestones, non-destructive examinations, and pressure testing.

Manufacturing Excellence

- **Precision Machining:** Maintain ± 0.01 mm tolerances across rotor alignment, sealing faces, and bearing housings via CNC machining for optimal reliability and interchangeability.
- **Welding & NDT:** Qualify welding procedures to ASME Section IX. Apply robust NDT techniques: RT for pressure boundaries, MPI for load-carrying welds, and DPI for finish surfaces.
- **Surface Protection:** Engineer internal surface protection (e.g., tungsten carbide, Xylan) for erosion/corrosion resistance and external coatings per ISO 12944-C5M for harsh climates.
- **Assembly & Integration:** Cleanroom assembly for internals; pre-calibrated rotor/seal pairs; HMI/VFD panels pre-configured and I/O validated.

- **Factory Acceptance Testing (FAT):** Simulated multiphase conditions shall be created using water and compressed air or nitrogen. A separator tank system will be employed to regulate GVF and recycle test fluids efficiently. The test bench shall be equipped with:
 - Flow meters: Coriolis, ultrasonic, or V-cone types.
 - Pressure transmitters: Installed at suction and discharge points.
 - Differential pressure sensors: Across the pump body.
 - Temperature sensors: Monitoring oil, bearing, and process fluid temperatures.
 - Vibration sensors: Mounted on motor and bearing housings.

The test setup must replicate the final field configuration of Control System Integration including:

- Full integration with VFD, PLC, and HMI.
- Validation of auto-trip functions, interlocks, and manual override capabilities.

A comprehensive test procedure shall be developed in alignment with project specifications and relevant standards such as API 676 (for positive displacement pumps) or API 610 (for centrifugal pumps). The procedure must include:

- Defined acceptance criteria and performance benchmarks.
- A detailed test matrix covering all operational scenarios.
- Embedded safety protocols and risk mitigation measures.
- A step-by-step execution plan for factory and site testing.

The following documents are required to support test execution and traceability:

- P&ID of the test setup.
- Datasheets for pump, motor, and VFD.
- Seal and bearing specifications.
- Calibration certificates for all instrumentation.
- Material and welding traceability records.

Prior to execution, a Pre-Test Meeting must be conducted involving the client, OEM, and third-party witness. This review ensures:

- Verification of instrument calibration certificates.
- Validation of the data acquisition system and control logic.
- Confirmation of readiness across mechanical, electrical, and instrumentation domains.
- **Packing & Logistics:** Crate and vacuum-seal internals, individually box rotor assemblies with VpCI protection, and mark lifting/CoG points with QR/barcodes for real-time tracking. Apply rust inhibitors; verify completeness at dispatch with shock/tilt indicators and preservation inspection forms.

- Delivery & Site Coordination:** Conduct transportation surveys for oversized loads, define modular breakdowns as needed, and clarify Incoterms with comprehensive insurance coverage. Coordinate delivery timelines with float for customs and site readiness; issue advanced notification for foundation and installation preparation.

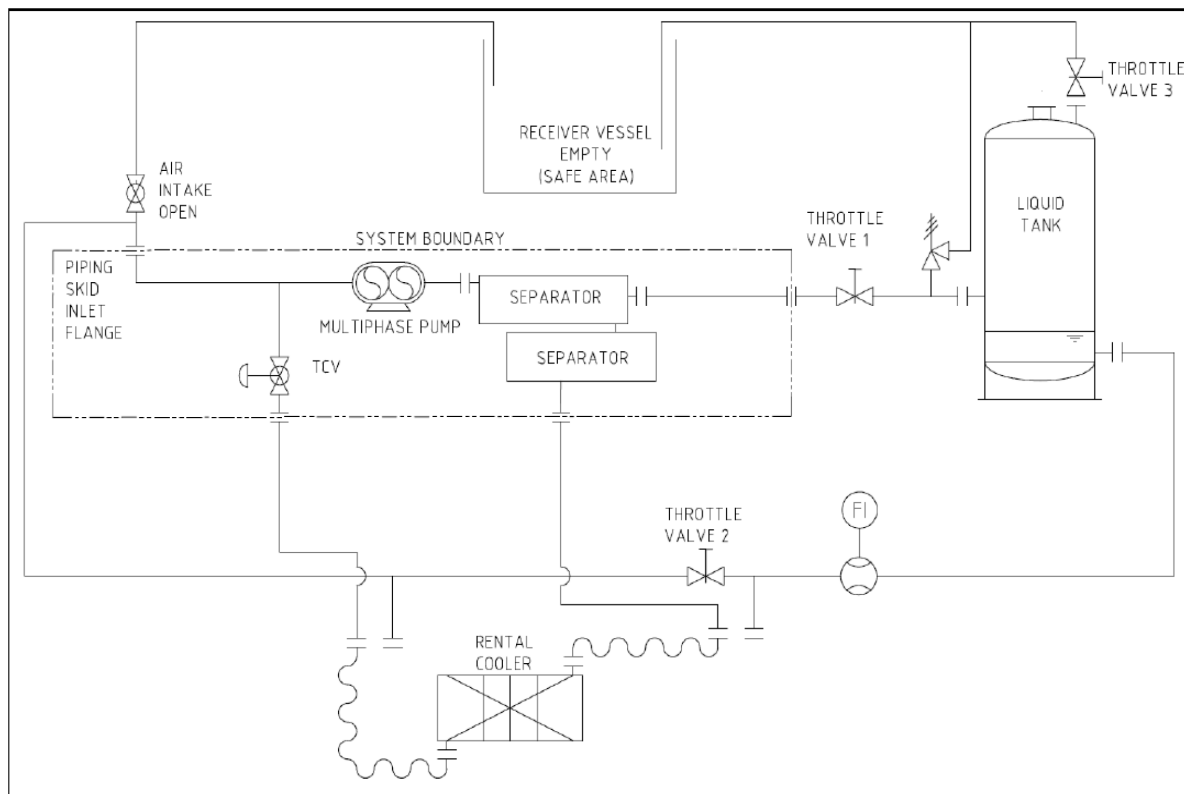


Figure 6—Test Setup for Multiphase Fluid Performance Evaluation

Site Support

- Site Receiving & Installation:** Enforce receiving inspection protocols against packing lists; mitigate storage risks (e.g., sun, sand ingress) and rotate shafts to prevent bearing deformation. Synchronize installation with vendor documentation and ensure cross-discipline collaboration (mechanical, electrical, controls).
- Field Commissioning Performance Test:** Verify integration with flowline, ESD system and Electrical System. Confirm actual pressure drop, power draw, and performance with reservoir fluids. Tests only after upstream/downstream facilities are stabilized. Monitor for issues caused by fluid instability, wax/sand, or inadequate suction pressure.
- Commissioning & Training:** Deploy OEM specialists for startup, alignment, and live parameter optimization; deliver operator training (classroom and field) on HMI, VFD, troubleshooting, and emergency response. Update as-built documentation to reflect field modifications and lessons learned.
- Post-Delivery Support:** Provide 12–24 month warranties covering parts and corrective actions; offer LTSAs with remote diagnostics and predictive analytics for uptime assurance.
- Lessons Learned & Digital Twin:** Institutionalize project closure reviews to capture deviations and integrate feedback into virtual models for advanced maintenance planning.

By adhering to these advanced protocols from procurement through post-delivery support, operators and EPCs can significantly enhance the safety, efficiency, and lifecycle performance of multiphase pump systems. The integration of digital tools and rigorous QA/QC at each stage ensures that systems meet the highest standards of reliability demanded by modern energy projects and aligns with ADIPEC's vision of operational excellence and future readiness.

Field-Proven Operational Best Practices for Reliable Multiphase Pumping Performance

Multiphase pumps (MPP) are increasingly adopted in upstream oil and gas production due to their ability to handle varying gas volume fractions (GVFs), fluctuating flow rates, and complex fluid compositions without requiring prior phase separation. However, successful deployment and sustained operation require adherence to rigorously defined operational protocols. This section outlines the best practices for field operation, monitoring, and lifecycle management of twin-screw MPP systems, drawing upon industry experience and operational data.

Pre-Commissioning and Startup Procedures

A robust startup sequence is critical to ensuring mechanical integrity and process safety. Prior to commissioning, operators should validate the following:

- **Alignment and Mechanical Verification:** Confirm precise alignment between the pump shaft and the driver (motor or turbine) using laser alignment tools. Improper alignment can lead to premature bearing or seal failure.
- **Seal and Barrier Fluid System Integrity:** Verify that the mechanical seal system is properly installed, and the barrier fluid reservoir is filled with clean, compatible fluid. Maintain the barrier fluid pressure at least 1–2 bar above the process pressure to prevent process ingress.
- **Instrumentation Calibration:** All field instruments, including pressure, temperature, and vibration sensors, must be calibrated and functionally tested to ensure accurate monitoring during dynamic operations.
- **Control System Testing:** Conduct comprehensive logic checks, including startup permissive, emergency shutdown (ESD), and interlock validations through the supervisory control system (ICSS or DCS/PLC). Verify that the variable frequency drive (VFD) is programmed with safe ramp-up profiles and torque limits.
- **Controlled Pressurization:** Initiate pump operation with a gradual increase in speed and differential pressure to ensure system stability and minimize mechanical stress.
- **Pipeline Purging:** Thoroughly flush all associated pipelines and pump internals to eliminate contaminants such as debris, welding residue, or hydrotest fluids prior to system start-up.

Steady-State Operating Guidelines

To maintain optimal performance and prevent operational upsets, the following parameters should be rigorously controlled:

- **Flow and Pressure Envelope Compliance:** Operate the pump within its design flow rate and differential pressure range. Twin-screw MPPs can tolerate 0–100% GVF; however, performance degradation is observed as GVF approaches 100%. If operated continuously at high GVFs, pump derating or auxiliary liquid injection may be required.
- **Suction Pressure Stability:** Sustained low suction pressure may lead to cavitation or gas lock. Maintain positive suction pressure by regulating upstream separators or buffer tanks and consider installing level control loops that modulate inflow to the pump. Operate above the pump's Net

Positive Suction Head Required (NPSHR) to preclude cavitation, utilizing pressure transducers and implementing interlock-driven shutdowns in the event of deviation.

- **Backpressure Regulation:** A discharge control valve or recirculation loop should be employed to maintain minimum discharge pressure and facilitate phase recondensation during high GVF events. This practice mitigates internal slippage and enhances volumetric efficiency.
- **Speed Modulation via VFDs:** Use speed control to adapt to fluctuating inlet conditions. Lowering pump speed during high GVF periods improves hydraulic balance and reduces vibration levels. Employ VFDs for dynamic speed regulation based on real-time suction pressure, flow rate, or motor torque, with closed-loop integration via SCADA or DCS platforms.
- **Lubrication and Thermal Management:** Deploy online viscosity and particle counters to monitor oil degradation. Schedule routine oil sampling and trend analysis in accordance with ISO 4406 cleanliness standards. Ensure uninterrupted coolant flow to motor and bearing assemblies, utilizing appropriate heat exchangers or air-cooled modules for high ambient temperature deployments.
- **Instrumentation and Automation:** Critical Parameter Instrumentation: Integrate redundant sensors for essential variables (suction/discharge/differential pressure, temperature, vibration, motor load, flow rate) and conduct periodic manual calibration checks. Establish tiered alarm logic (warning and trip) for anomalies such as pressure excursions, motor overload, or seal failure, with clearly defined escalation and response protocols.

Table 6—Key Operational Parameters and Recommended Ranges

Parameter	Recommended Range	Remarks
GVF (Gas Volume Fraction)	0–80% (max up to 95%)	Above 80% may cause lubrication issues
Suction Pressure	> NPSHR	Maintain to avoid cavitation
Vibration Level	< 5 mm/s RMS	Monitor for misalignment or bearing wear
Lubrication Oil Temp	60–80°C	High temp reduces oil life
Differential Pressure	As per pump design (e.g., 20–40 bar)	Maintain to stay in efficient zone
Motor Load (%FLA)	70–90%	Avoid overload conditions

Condition Monitoring and Predictive Diagnostics

Continuous condition monitoring enables proactive maintenance and extends equipment life. Recommended practices include:

- **Critical Parameter Surveillance:** Track real-time suction/discharge pressures, differential pressure, motor torque, bearing and seal temperatures, and vibration amplitudes. Sudden anomalies often precede mechanical failure.
- **Data Trending and Threshold Alarming:** Integrate the pump monitoring system with a SCADA or historian platform. Define alarm setpoints based on equipment baseline trends e.g., ±10% vibration increase or a 5°C rise in seal temperature may trigger early inspection.
- **Predictive Analytics Integration:** Employ machine learning models or digital twin simulations to correlate field conditions with potential failure modes, improving maintenance scheduling and resource allocation.
- **Pump Efficiency Assessment:** Regularly compare field efficiency (calculated from flow, pressure, and power data) to original equipment manufacturer (OEM) curves to identify performance degradation or misalignment.

- **Lifecycle Asset Planning:** Include rotor-stator wear evaluations as part of annual inspections. Leverage vendor support for rotor balancing, rebuild kits, and repair recommendations.
- **Operator Training and SOP Adherence:** Conduct simulator-based, hands-on operator training, with periodic refreshers emphasizing abnormal scenario management (e.g., seal failures, high GVF events). Maintain and routinely update standard operating procedures for all operating phases start-up, normal operation, shutdown, and troubleshooting incorporating lessons learned from incidents and near-misses.

Multiphase pumps offer flexible and cost-effective solutions for challenging upstream production scenarios, especially in marginal or remote fields. However, to fully realize their benefits, operations must be underpinned by disciplined execution of startup protocols, real-time condition monitoring, strategic maintenance planning, and digital performance optimization. The integration of such best practices leads to enhanced equipment availability, lower total cost of ownership, and improved production assurance.

Smart Maintenance Strategies for Multiphase Pumps: Enhancing Uptime and Performance

Multiphase pumping systems face challenges such as **erosive wear, gas locking, and seal degradation** due to fluctuating pressures, abrasive particulates, and corrosive media. A proactive maintenance regime is essential to mitigate these risks while maintaining operational efficiency. This section consolidates field-proven methodologies for sustaining multiphase pump reliability. The maintenance philosophy has been established as a hybrid approach, combining condition-based and risk-based methods.

Condition Monitoring & Predictive Maintenance

Continuous monitoring of key indicators enables early detection of wear and emerging failures. Recommended techniques include:

- **Vibration Analysis:** Identifies bearing degradation, rotor imbalance, and misalignment through high-frequency data trending.
- **Infrared Thermography:** Detects abnormal thermal signatures in bearings, gearboxes, and mechanical seals.
- **Lubricant Analysis:** Oil sampling and analysis detect particulate contamination, water ingress, and additive depletion (as per ISO 4406 cleanliness codes).
- **Performance Trending:** Deviation in discharge pressure, flow rate, or power draw often precedes mechanical deterioration.
- **Motor Current Signature:** Analyzes the electrical current drawn by motors to detect mechanical and electrical faults.

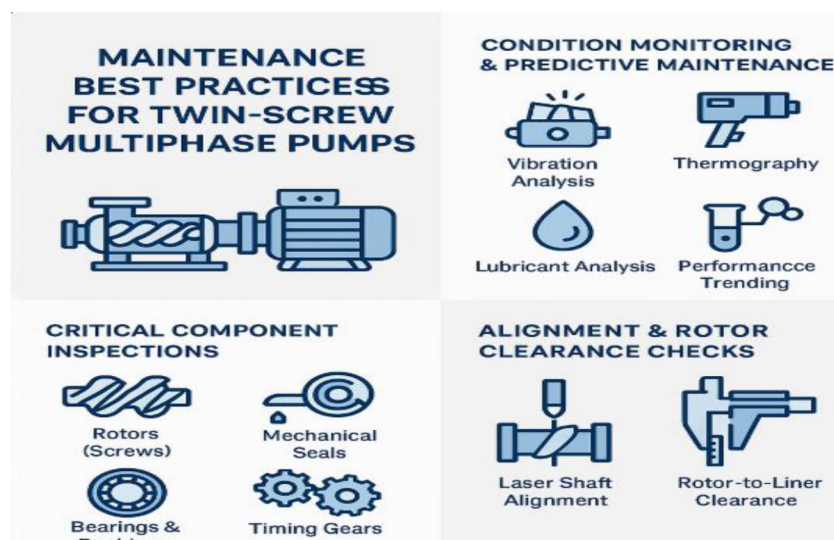


Figure 8—Illustrates core condition monitoring methods for twin-screw MPPs.

Critical Component Inspection Schedule

Inspection and shutdown planning typically follows a structured approach combining periodic checks and major overhauls. Minor inspections, conducted every six months, focus on verifying alignment, instrumentation accuracy, mounting bolt integrity, and cable terminations. Major overhauls, scheduled every 2 to 4 years, involve more intensive procedures such as rotor removal with non-destructive testing (NDT), replacement of bearings and seals, and pressure testing to benchmark performance. Turnaround planning is strategically timed during low production periods to minimize operational impact and ensure resource availability. Key rotating and static components must be inspected periodically based on severity of service and operational hours. The table below summarizes recommended inspection intervals.

Table 7—Recommended Inspection Intervals for Critical MPP Components

Component	Inspection Focus	Frequency
Rotors (Screws)	Erosion, corrosion, coating degradation	6–12 months
Mechanical Seals	Leakage, flush fluid contamination	Quarterly
Bearings & Bushings	Clearance, lubrication film integrity	Monthly (vibration-led)
Timing Gears	Backlash, tooth wear, lubrication consistency	Annually
Component	Inspection Focus	Frequency
Motor Load (%FLA)	70–90%	Avoid overload conditions
Lube oil system	Oil level/top-up, filter change	Weekly / Quarterly
Seals & bearings	Visual inspection, leak check	Monthly / Quarterly
Vibration probes	Calibration, verification	Quarterly
Control system (VFD, PLC)	Software updates, I/O validation	6–12 months
Safety devices	ESD, alarms, PSV functional test	Annually

Lubrication Management

Proper lubrication is essential to avoid contact fatigue, thermal buildup, and wear:

- Use **synthetic lubricants** with enhanced oxidation resistance for high-temperature or sour gas applications.

- Integrate **automatic lubrication systems** to ensure continuous supply under varying load conditions.
- Maintain oil cleanliness per **ISO 4406:1999 standards** (target $\leq 18/15/13$ for critical equipment).

Mechanical Seal System Optimization

Mechanical seal failure is a common cause of unplanned shutdowns in MPP systems:

- Use **pressurized dual seal systems** with barrier fluids suitable for high-GVF applications.
- Ensure **API 682-compliant flush plans** are implemented with adequate flow rates and redundancy.
- Install **dual filtration** in the seal flush line to prevent abrasive particles from reaching sealing surfaces.

Alignment & Rotor Clearance Verification

Rotor dynamics and pump efficiency are heavily influenced by shaft alignment and internal clearances:

- Conduct **laser shaft alignment** during installation and major maintenance (target misalignment: <0.05 mm).
- Verify **rotor-to-liner clearance** against OEM specifications during overhauls. Excessive clearance leads to slippage and efficiency loss.

Failure Mode Analysis & Mitigation

Reliable operation requires a comprehensive understanding of potential failure modes and proactive mitigation strategies.

Understanding the possible failure modes helps in developing a robust maintenance and monitoring plan. Identifying and addressing root causes prevents recurring failures.

Table 8—Representative examples for Common failure modes in multiphase pumps

Failure Mode	Probable Root Cause	Mitigation Strategy
Seal leakage	Contaminated barrier fluid	Dual filtration in seal system
Bearing degradation	Inadequate or improper lubrication	Automated grease injection with alarms
Rotor coating erosion	Sand or scale ingress	Use of hard-faced rotors; upstream filtration
Gear tooth wear	Misalignment or oil contamination	Real-time oil monitoring and backlash checks

By thoroughly analyzing the failure modes and implementing targeted mitigation strategies, operators can significantly extend the service life of multiphase pumps, reduce unplanned shutdowns, and ensure safe, efficient operation in demanding environments.

Key considerations from current twin screw pump installations include ensuring the inlet header is designed for even distribution of liquids and gas over the pump skids. The lube oil system and seal oil system should use separate oil tanks. Mechanical seals are to be double pressurized types, with API plan 53 as a minimum requirement.

Development Case Study: Strategic Deployment of Multiphase Pumping Systems in Haliba Field Development

The deployment of the reliability-by-design framework in Haliba Field multiphase pump (MPP) installations **yielded measurable improvements in system performance, reliability, and cost efficiency**. Implementation of this framework at Al Dhafra yielded significant improvements:

- The Haliba Central Gathering Facilities (CGF) implemented Multiphase Pumps (MPPs) to transfer untreated well fluids approximately 65 km to ADNOC Onshore's ASAB CDS for processing. Using **MPPs accelerated early production** and provided **pressure boosting**.
- The multiphase pump system demonstrated exceptional reliability and operational performance. **Availability consistently exceeded 95%**, supported by a **Mean Time Between Failures (MTBF)** greater than **4000 hours**, highlighting the robustness of both design and maintenance strategies.
- **Preventive Maintenance (PM) compliance** was maintained at **≥95%**, ensuring timely interventions and sustained asset health.
- **Unplanned downtime** remained below **2% of total run hours**, minimizing production disruptions and enhancing throughput.
- **Vibration levels** were controlled at or below **5 mm/s RMS**, in accordance with **ISO 10816-3**, safeguarding mechanical integrity and reducing wear across critical components.
- Additionally, **Cost Savings**: Reduced maintenance costs through predictive monitoring and optimized material use.

Conclusion

Drawing upon the extensive field experience from Al Dhafra Petroleum's operations, particularly in the Haliba Field, the study establishes a robust, field-tested methodology that transcends conventional approaches to multiphase pumping.

Central to this framework is the integration of systematic selection criteria, reliability-centered design, advanced manufacturing protocols, and best-in-class operational and maintenance practices. By rigorously aligning pump selection with actual field operating conditions accounting for variables such as gas volume fraction, viscosity, solids content, and fluctuating production profiles operators can significantly mitigate risks of premature failure, inefficiency, and unplanned downtime. The comparative analysis between twin-screw and helico-axial pump technologies provides clear guidance for selecting the optimal system architecture based on application-specific needs.

A key contribution of this work is the detailed engineering and safety strategy, encompassing hydraulic and mechanical integrity, high-fidelity instrumentation, and integrated control logic. The manuscript underscores the critical importance of comprehensive performance testing, adherence to international quality and safety standards, and the adoption of advanced manufacturing and qualification protocols. These measures collectively ensure that multiphase pump installations meet or exceed the demanding requirements of remote, marginal, and high-GOR field environments.

Operational excellence is further reinforced through the establishment of best practices in startup, steady-state operation, condition monitoring, and maintenance. The emphasis on predictive maintenance, digital integration, and failure mode mitigation empowers operators to extend equipment lifespan, minimize lifecycle costs, and maintain asset health even under harsh service conditions. The case study from the Haliba Field validates the practical efficacy of the proposed framework, demonstrating tangible improvements in pump availability, mean time between failures, cost savings, and overall production assurance.

In summary, this manuscript provides a holistic, actionable blueprint for upstream operators aiming to maximize the value of multiphase pumping technology. By adopting this structured methodology, organizations can enhance production efficiency, operational reliability, and cost-effectiveness across a diverse range of field scenarios. The lessons and strategies detailed herein are scalable and adaptable, offering a pathway for continual improvement in the face of evolving energy sector demands and technical complexities. Future work can build upon this foundation by incorporating emerging digital solutions, advanced analytics, and further field validation to refine and extend the benefits of optimized multiphase pumping systems.

References

Execution and Insights from Haliba Field Development: Deployment of Multi-Phase Pump Installations
API 610 - Centrifugal Pumps
API 676 - Positive Displacement Pumps
ISO 10816-3 - Vibration Severity Standards
API RP 682 - Pump Sealing Systems
ISO 21049 - Petroleum Pump Safety Standards